

Project Planning & Timeline

Development Milestones

The implementation timeline was planned across four distinct sprints:

Sprint 1: Data Preparation

Gathering meteorological spreadsheet logs. Capping precipitation anomalies using Interquartile Range (IQR).

Sprint 2: Model Training

Fitting StandardScaler weights. Evaluating Decision Tree, Random Forest, KNN, and XGBoost classifiers. Decision Tree yields 96.55% accuracy.

Sprint 3: Web Application Implementation

Designing templates with Bootstrap 5. Implementing Flask routing, AJAX form validation, and inline step loaders.

Sprint 4: Deployment & Tests

Creating automated route unit tests. Configuring vercel.json. Committing to GitHub and deploying to Vercel.